INTER-UNIVERSITY GENAI HACKATHON FOR SDGS 2026

# Resilience Travel Assistant

## TEAM 1076 / TOMORROW

**Green Transport & Smart Mobility**

A travel decision-support tool that helps Hong Kong and cross-border commuters choose departure times, compare disruption risks, and prepare alternatives using transport data, transparent rules, and constrained explanations.



**CLEARER CHOICES UNDER DISRUPTION**


**TEAM MEMBERS**

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# Brand Identity & Visual Design



**Resilience Travel Assistant**
**Know your route. Know when to go.**

## / LOGO DESIGN

- Multiple Routes, Clear Decisions

Interconnected routes represent alternative travel options.

## / VISUAL IDENTITY

- Calm, Clear & Reliable

Deep teal and dark blue create a calm, trustworthy visual identity.

## / PRODUCT PHILOSOPHY

- Fixed Commute. Transparent Rules.

Data-driven insights help commuters compare risks and plan alternatives.

# A disruption changes the whole journey

**11.7 million**

public transport journeys per day
in Hong Kong, 2024

**158.1 mm**

rainfall in one hour at HKO
7 September 2023, 23:00–24:00

A traveller still has to decide whether the entire trip is feasible, how much time to allow, and when an alternative stops being useful.

Sources: Hong Kong Transport Department (2025 digest), Hong Kong Observatory (September 2023)

# The first users have a deadline




## Fixed commute, visible constraints

A traveller enters an origin, destination and latest arrival time, then sets limits for extra cost and transfers.

The same screen can switch between normal, rain and transport-disruption fixtures.

The first target users are students and workers with repeated trips and a hard deadline. Interviews and field trials remain pending.

# A working route comparison




| Recommended: Lok Ma Chau | Alternative: Lo Wu | Alternative: Shenzhen Bay |
|---|---|---|
| 81 min + queue and buffer | 91 min + queue and buffer | 132 min + queue and buffer |
| Arrival 09:33–09:59<br>Low risk / Medium confidence | Arrival 09:49–10:31<br>Medium risk / Low confidence | Arrival 10:32–11:17<br>Medium risk / Low confidence |
| | | Conservative ETA misses deadline |

Actual local prototype screenshot, 17 October 2026. Fixed demonstration data. English translation below.

# The response changes with the scenario

## Normal




08:09 · high confidence

## Heavy rain




07:58 · medium confidence

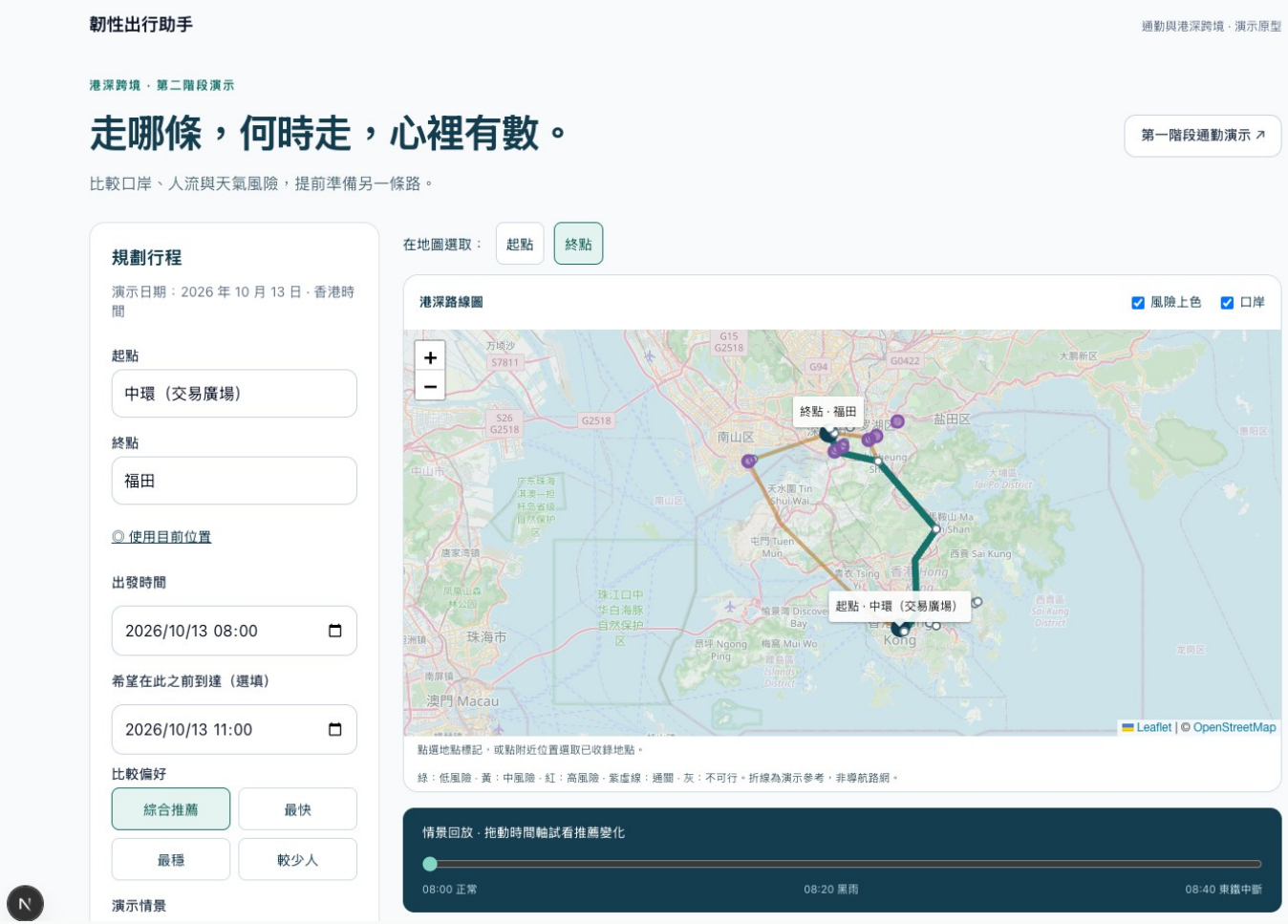
## Transport disruption




07:22 · low confidence

Sources: archived browser QA captures (13 Oct 2026) and release recheck (17 Oct 2026).

# Two routing modes with explicit boundaries




## Actual second-stage page

Trip form, map, risk overlay and scenario replay are visible.

9,466 stops and 2,452 usable routes in the HK snapshot.

The public Shenzhen leg requires a private server-side Amap key. Static timetables are not live ETAs.

Source: integrated app README, transit data and recorded provider probes.

# GenAI explains a structured decision

### WHAT THE MODEL DOES

With a private server-side key, qwen-plus turns selected decision facts into a short travel explanation.

### WHAT THE MODEL CANNOT CHANGE

Routes, departure times, ETAs, risk labels, and go / hold decisions come from the rule-based engine.

**SAFEGUARDS** Only whitelisted facts are sent; unsupported numbers are rejected. Timeout, error, or missing key leaves the structured decision intact.

Implemented in the three application APIs. Local keyed use has been observed; a fresh public GitHub checkout contains no credentials.

**Model adapter and validation code** github.com/jijie114514-oss/eight-university-hackathon

Source: src/lib/ai/explain.ts and ai-explanation.test.ts. Model availability depends on server configuration.

# A tested prototype, with evidence limits

**595**

tests passed · integrated app
43 test files

**1,129**

passes across three saved versions
Includes shared regressions

Lint, type checks and production builds passed. No completed human usability study is claimed.




**Browser QA · 390px**

Source: fresh vitest run, 18 Oct 2026; release-checks/RELEASE_REVIEW.md for earlier review.

# Impact will be measured in a small pilot

## SDG 11.2

Support informed use of public transport within a traveller’s time and cost constraints.

## SDG 13.1

Help users recognise disruption and defer travel when the scenario cannot support a recommendation.

## Proposed pilot

10–15 consenting commuters over 5–10 travel days. Compare with their existing planning tools.

## Measures and guardrails

Planning time, lateness, extra cost and understanding of uncertainty. No measured carbon or congestion impact yet.

SDG alignment: UN targets 11.2 and 13.1. Pilot targets are proposed, not achieved outcomes.

# A focused path to adoption

## Initial access

A mobile web app for one university or employer commute community. A partner is not yet confirmed.

## Funding hypothesis

Free core traveller access, with an institutional pilot or support contract if value is demonstrated.

## Operating costs

Node hosting, address and route providers, feed updates, monitoring and a managed foot router.

## Expansion gate

Expand corridors only after coverage, usability and provider costs meet agreed pilot thresholds.

Business model and costs are hypotheses for validation. No customers or institutional partners are confirmed.

# Uncertainty remains visible to the user

## Travel advice

Show sources and uncertainty. No calibrated on-time probability, live queue guarantee or emergency navigation claim.

## Privacy

Server-side API keys. Search and routing can send locations to external providers. A pilot needs consent and retention controls.

## Inclusion

Mobile layout checks passed. Screen-reader, step-free routing and real-device testing still need validation.

## Maintenance

Production dependency audit: zero known findings at review. Seven high development-tool findings remain disclosed.

Source: implementation and release review. Controls reduce risk but do not certify route safety.

# The next milestone is a measured pilot

| Period | Planned milestone | Evidence required |
|---|---|---|
| Weeks 1–2 | Deploy and verify a bounded corridor | Provider probes, accessibility review and consent flow |
| Weeks 3–4 | Run user tasks and explanation comparisons | Task completion and factual-error analysis |
| Weeks 5–6 | Run the commute pilot | Planning time, lateness and extra-cost comparison |

**We seek a pilot host, consenting commuters and transport-data review.**

Proposed plan after the competition. Milestones depend on user recruitment and provider access.

# Appendix  A reproducible prototype

https://github.com/jijie114514-oss/eight-university-hackathon

Live demo: https://eight-university-hackathon.vercel.app
Local: Node.js 24 · node deploy-local.mjs integrated
Second stage: /  ·  First stage: /phase1

Select “Central to Futian” for the fixed demo.
Switch normal service, East Rail disruption and black rainstorm.
Open /phase1 for the first-stage commute.

Release checks and known limitations

The public repository, live demo and embedded screenshots provide prototype proof.

# Appendix  Recorded provider coverage

| Recorded probe | Result | Interpretation |
|---|---|---|
| HK district-coordinate samples | 12 / 12 | Reference route to Shenzhen Bay HK bus terminal |
| Initial complete cross-border samples | 3 / 3 | Central, Tsim Sha Tsui and Mong Kok to Futian |
| Shenzhen named POI samples | 8 / 8 each direction | One POI per district with Central as HK endpoint |

The public GitHub repository contains no provider key. Saved probes are not field trials or an all-city coverage guarantee.

Archived probes dated 17 October 2026. Reviewed from supplied artifacts, not re-run with a live key in this release.

# Appendix  WorkBuddy evidence

## Documented contribution

A decision-interface draft with rules for provenance, missing information and constrained explanation.

## Human integration decision

Replace the placeholder corridor, reconcile field naming and add supported fields to the main contract.

## Reusable asset

A schema-and-validation workflow that another team can apply to a source-backed decision tool.

## Attribution boundary

No WorkBuddy token counts, speed gains or underlying model identity are claimed. Other tools’ work is excluded.

Source: archived WorkBuddy contract and integration reconciliation. Separate optional report prepared.

# Appendix  Sources and evidence

Competition submission requirements

Judging criteria

Hong Kong public transport patronage

September 2023 rainfall and disruption

UN SDG 11 transport access

UN SDG 13 climate resilience

Release verification and limitations

Transit coverage and probe definitions

Explanation interface and safeguards

Archived WorkBuddy contract

Human integration change record

Links are embedded. Full URLs and source context also appear in speaker notes.